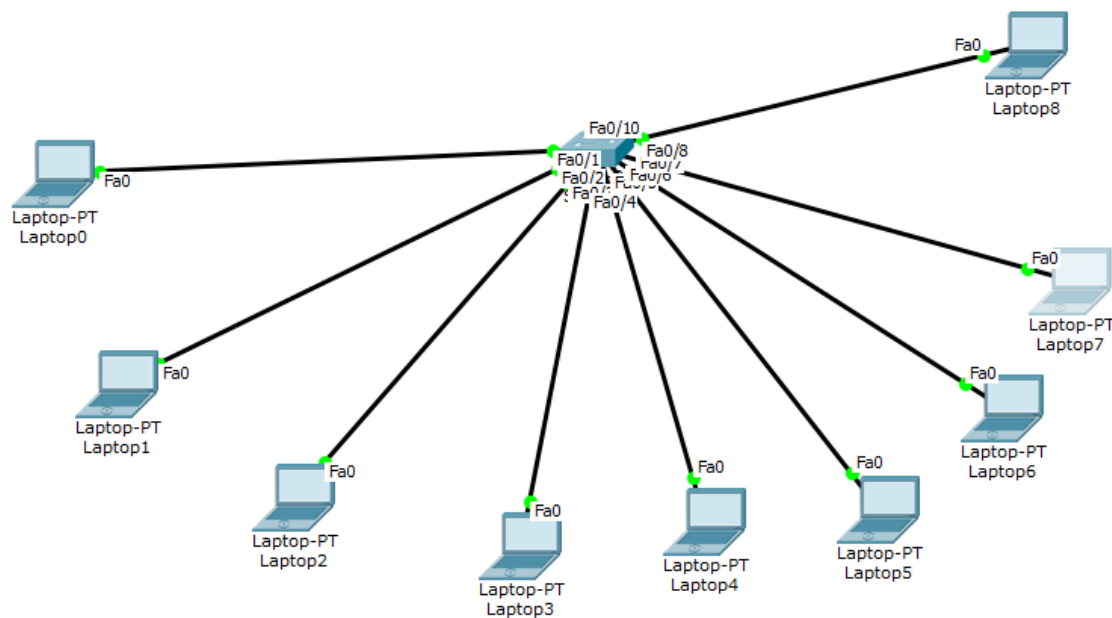




Switch0

Physical Config CLI

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up

Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range Fa 0/1-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#switchport port-security violation shutdown
Switch(config-if-range)#exit
Switch(config)#
```

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```
interrace range not validated - command rejected
Switch(config)# interface range fa0/9-24
Switch(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
Switch(config-if-range)#
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to down

Switch(config-if-range)#exit
Switch(config)#
```

Switch0

Physical Config CLI

IOS Command Line Interface

```
Switch(config)# interface g0/1
%Invalid interface type and number
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	up	up
FastEthernet0/6	unassigned	YES	manual	up	up
FastEthernet0/7	unassigned	YES	manual	up	up
FastEthernet0/8	unassigned	YES	manual	up	up
FastEthernet0/9	unassigned	YES	manual	administratively down	down
FastEthernet0/10	unassigned	YES	manual	administratively down	down

--More--

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Show port-security

Switch0

Physical Config CLI

IOS Command Line Interface

```
FastEthernet0/18      unassigned      YES manual administratively down down
FastEthernet0/19      unassigned      YES manual administratively down down
FastEthernet0/20      unassigned      YES manual administratively down down
FastEthernet0/21      unassigned      YES manual administratively down down
FastEthernet0/22      unassigned      YES manual administratively down down
FastEthernet0/23      unassigned      YES manual administratively down down
FastEthernet0/24      unassigned      YES manual administratively down down
Vlan1                 unassigned      YES manual administratively down down
Switch#show port-security
```

Secure	Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
	Fa0/1	1	0	0	Shutdown
	Fa0/2	1	0	0	Shutdown
	Fa0/3	1	0	0	Shutdown
	Fa0/4	1	0	0	Shutdown
	Fa0/5	1	0	0	Shutdown
	Fa0/6	1	0	0	Shutdown
	Fa0/7	1	0	0	Shutdown
	Fa0/8	1	0	0	Shutdown

Switch#

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Detailed view

Switch0

Physical Config CLI

IOS Command Line I

```
FastEthernet0/21      unassigned      YES manual administratively down down
FastEthernet0/22      unassigned      YES manual administratively down down
FastEthernet0/23      unassigned      YES manual administratively down down
FastEthernet0/24      unassigned      YES manual administratively down down
Vlan1                 unassigned      YES manual administratively down down
```

Switch#show port-security

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Fa0/1	1	0	0	Shutdown
Fa0/2	1	0	0	Shutdown
Fa0/3	1	0	0	Shutdown
Fa0/4	1	0	0	Shutdown
Fa0/5	1	0	0	Shutdown
Fa0/6	1	0	0	Shutdown
Fa0/7	1	0	0	Shutdown
Fa0/8	1	0	0	Shutdown

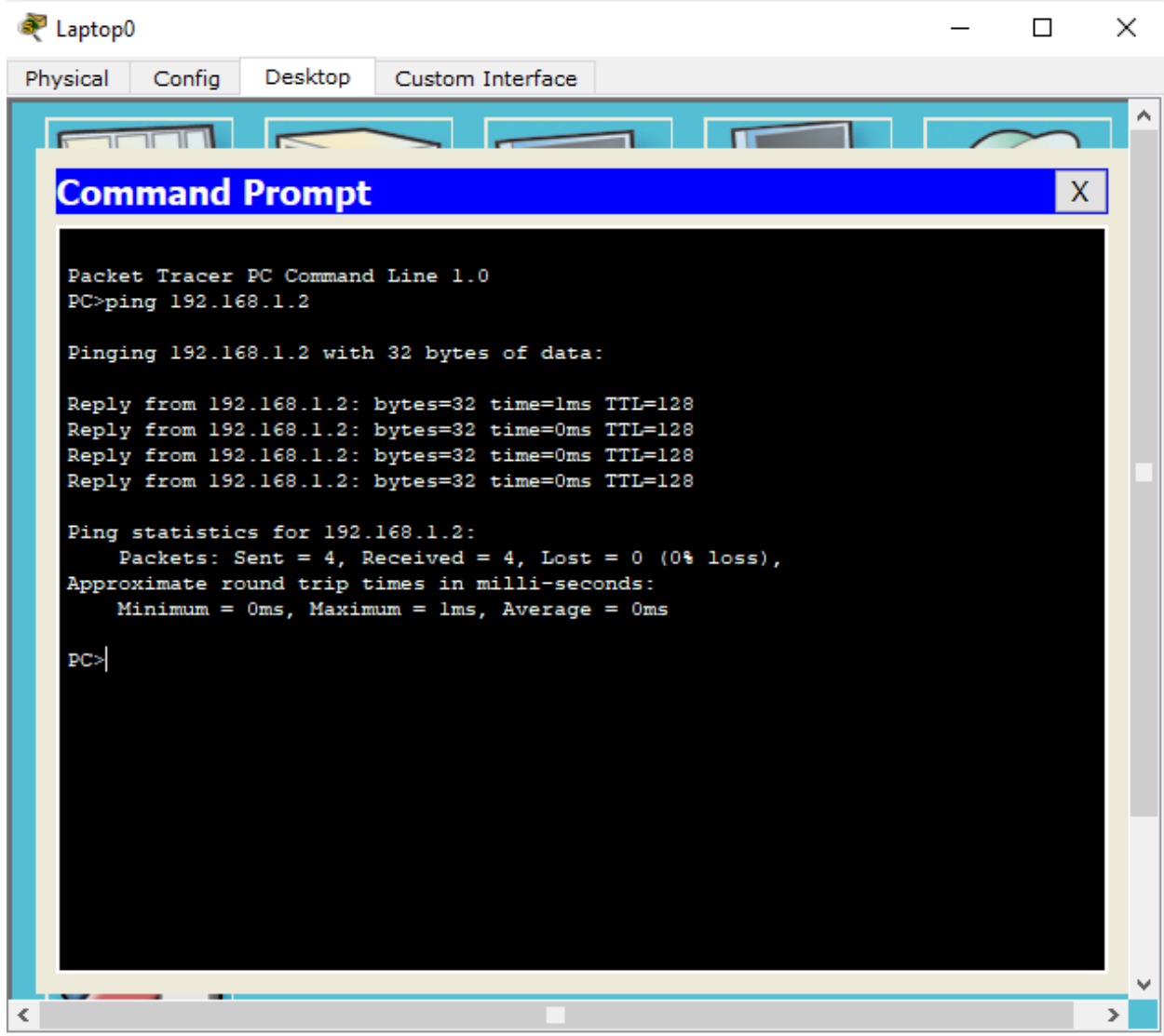
Switch#show port-security interface Fa0/1

```
Port Security      : Enabled
Port Status        : Secure-up
Violation Mode     : Shutdown
Aging Time         : 0 mins
Aging Type         : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 0
Configured MAC Addresses : 0
Sticky MAC Addresses : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

Switch#show port-security interface Fa0/3

```
Port Security      : Enabled
Port Status        : Secure-up
Violation Mode     : Shutdown
Aging Time         : 0 mins
Aging Type         : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 0
Configured MAC Addresses : 0
Sticky MAC Addresses : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
```

Switch#



The screenshot shows a Packet Tracer desktop environment. At the top, there are tabs for 'Physical', 'Config', 'Desktop', and 'Custom Interface'. The 'Desktop' tab is active, displaying a virtual desktop with several icons. A 'Command Prompt' window is open in the center, titled 'Command Prompt' with a close button (X). The text inside the window is as follows:

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=0ms TTL=128
Reply from 192.168.1.2: bytes=32 time=0ms TTL=128
Reply from 192.168.1.2: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>|
```

Switch0

Physical Config CLI

IOS Command Line Interface

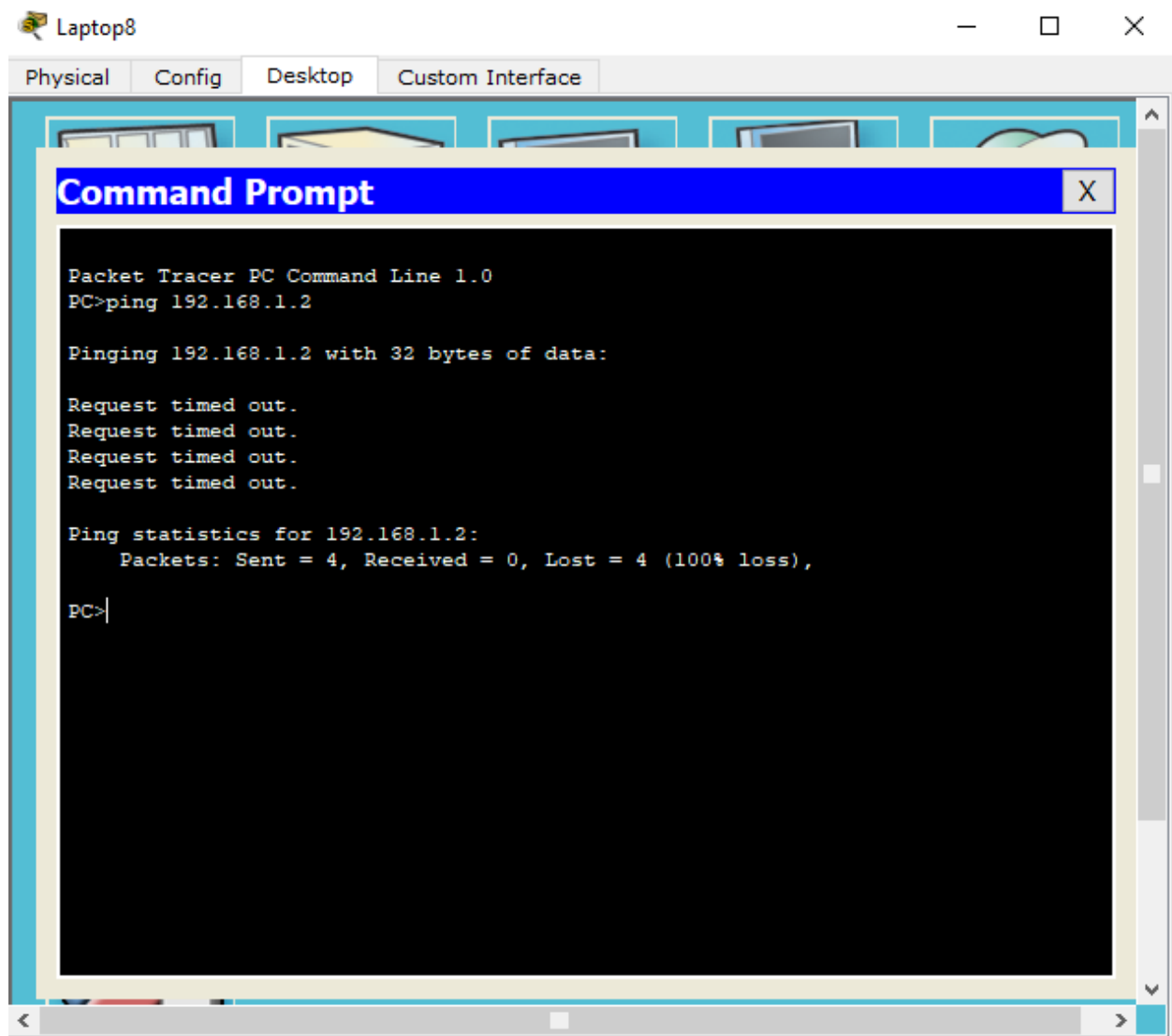
```
Switch>en
Switch# show port-security address
```

Secure Mac Address Table			
Vlan	Mac Address	Type	Ports
Remaining	Age		
(mins)			
1	0090.2B35.68E9	SecureSticky	-
1	000A.4175.3796	SecureSticky	-
1	00D0.FFD7.6996	SecureSticky	-
1	000A.F32E.5E78	SecureSticky	-
1	0060.2F83.6DD2	SecureSticky	-
1	0060.706C.2301	SecureSticky	-
1	0001.4397.9965	SecureSticky	-
1	0030.A345.BBC6	SecureSticky	-

```
-----
Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#
```

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From the attacker side



After Violation of attacker side

 Switch0

Physical Config CLI

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
down
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to
down

Switch#show port-security interface Fa0/1
Port Security           : Enabled
Port Status             : Secure-shutdown
Violation Mode          : Shutdown
Aging Time              : 0 mins
Aging Type              : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Last Source Address:Vlan : 000D.BD10.4D34:1
Security Violation Count : 1

Switch#
```

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Switch0

Physical Config CLI

IOS Command Line Interface

```
down
Switch#show port-security interface Fa0/1
Port Security          : Enabled
Port Status            : Secure-shutdown
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 000D.BD10.4D34:1
Security Violation Count : 1

Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/1      1              1              1          Shutdown
Fa0/2      1              1              0          Shutdown
Fa0/3      1              1              0          Shutdown
Fa0/4      1              1              0          Shutdown
Fa0/5      1              1              0          Shutdown
Fa0/6      1              1              0          Shutdown
Fa0/7      1              1              0          Shutdown
Fa0/8      1              1              0          Shutdown
-----
Switch#
```

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